Source audit — README + docs/00–07 vs. ledger and raw sources

Audited: 2026-10-24, workspace clock ~00:52. No network; only local files. Snapshot: `README.md` (mtime 00:51:56), `docs/00`–`docs/07`, `docs/SOURCES.md`, `docs/_research/{harness-facts,notebook-facts}.md`, plus the two raw sources `harness_readme_raw.md` (627 lines) and `getting_started_notebook.txt` (615 lines). `README.md` was being edited during the audit; findings reflect the 00:51:56 revision.

Method:

1. Every `S5:<n>` / `S6:<n>` reference in the corpus (184 total) was expanded and compared line-by-line with the raw files.
2. Every `[S#]` was checked for a matching `## S#` entry; ledger facts were compared with the sentence that cites them.
3. Arithmetic and statistics in the docs were recomputed (12 h projections, prize sums, per-task score steps, sign-test p-values).
4. Claims were classified against the ledger's own rules: official beats community; single community runs need `[COMMUNITY]`; planning numbers need `[ESTIMATE]`.

What passed: all 49 source IDs resolve; no `S5`/`S6` line reference is out of range and none points at an unrelated section except the two README format examples (#33); the sign-test p-values in 06 §3.3 (0.125 / 0.109 / 0.021) recompute exactly; the 12 h projections (5 min→10 h, 5.5→11 h, 8→16 h at N=120) are correct; prize sums ($37k$+18k+$10k$ =65k; $65k$+35k+$10k$ =110k) are correct; the absence of a 12 h rule in `harness_readme_raw.md` is confirmed.

Counts: 3 Critical, 7 High, 17 Medium, 13 Low (40 findings).

Findings (prioritized)

#	Sev	File:section	Claim	Problem	Source that should be used / `UNSUPPORTED`	F
1	**Critical**	README Status + §1.3 + §4 Path A/B; 02 §2, §2.3; 03 intro; 05 R12	"A public prompt-only configuration already reaches 0.24 public"; "the public evidence says a prompt-only 31B agent can reach 0.24"; "Prompt-only baseline is already strong (0.24 public)"	The ledger's own S36 entry says the 0.24 is *author-reported*, that the page sidebar shows "Best Score 0.18 V3", and that the notebook "does not run inference". The body text states it as an established result. 02 §2.3's rationale bullets are all from S38 (the 0.18 notebook), so the "why this config works" evidence is attached to a different artifact. README was since hedged; 02/03/05 were not.	S36 (with its caveat), S37, S4, S38	C i r 0 o r § e (a r

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2	**Critical**	02 §2.1, §2.3, §10; README §7	"The 240 s command timeout is also used by the scoring pytest run … a shorter timeout can fail a correct patch"	Contradicts the official harness: Phase-2 pytest is **fixed at 300 s on the scorer** and has no `eval_config.yaml` key (`harness_readme_raw.md:506`). `evaluation.timeout_seconds` only sets the Phase-1 command timeout (`:505`). So the recommended 240 s cannot affect verification at all, and the stated failure mode cannot occur. 01 §8.1 states the correct 300 s, so the corpus contradicts itself.	S5:505-506 (official) vs S38 (community, contradicted)	R "; t P S w r ti p
3	**Critical**	README §4 Path A vs §9; 02 §2, §10 vs 05 R1	Three different "default" budgets: 5 min/24 calls/thinking 2048 (README §4), 28 calls/8 min (README §9, 02 §10), 5–6 min (05 R1), 8 min/28 calls/4096 as "best starting point" (02 §2)	README §9 still says "respects the 28-call/8-minute budget" after §4 was changed to 5/24; 02 §2 and §10 still present 8/28/4096 as the baseline to reproduce. A reader cannot tell which config to ship, and the choice changes the global-time projection.	S36/S38 for the reference config; S29/S34 for the time projection; no source mandates either default	M c s 8 c a § a
4	High	00 §3; SOURCES E6	"at about 120 hidden tasks, one task is about 0.0083"	Final standings are computed on the **private** half (~60 tasks, S4 + S39), where one task is ≈1/60 ≈ 0.017. 0.0083 is the blended 1/120 figure and never appears on the leaderboard being optimised. The docs use it to size noise ("±2 tasks"), which understates the effect on the final score by 2×.	S4 (50/50 split), S39 ("public split ~58 tasks; private ~60"), S2 (~120 total)	S a k " th n
5	High	02 §1.3, §1.5	"the public 0.18 analysis found 7 of 8 timeouts came from the first stage in a two-stage test (S28)"; "`exit_loop` is not in the tool registry (S28)"	S28 is the harness-issues thread (compaction, LoRA KV, tool-output bugs). Neither fact is in the S28 entry, and the "public 0.18 analysis" is S38, not S28. The `exit_loop` fact is actually supported by the tool list in S5:341.	`exit_loop`: S5:341. 7-of-8 timeouts: UNSUPPORTED (no ledger entry)	C r s p e a
6	High	03 §9.1; 04 §3.2	"Use the starter notebook's config: `max_loras=1`, `max_lora_rank=16`"; `--max-loras 1 --max-lora-rank 16` presented as "essential flags"	The scorer derives these: `max_loras = max(1, len(adapter_map))`, `max_lora_rank = max(detected_ranks) else 128` (S5:153, S6:283-284). Hard-coding 16 silently under-serves any rank>16 adapter, so the local A/B is not the scorer's stack — exactly the parity failure the same paragraph warns about. 01 §4.1 states the derivation correctly.	S5:153, S6:283-284	S c c a o a

#	Sev	File:section	Claim	Problem	Source that should be used / UNSUPPORTED	F
7	High	03 §4.1, §4.4, §5.1; 05 R7	"used five temperatures"; "verified the token stream was identical … (12,602 tokens in both)"; "thought blocks 2.5k–14.7k chars"	None of these three numbers appears in the S25 entry. "Five temperatures" also contradicts 04 §6.1, which lists four (0.2/0.5/0.8/1.0). These are the only quantitative anchors for the thinking-mismatch lesson.	S25 — must be extended with the exact quotes, else UNSUPPORTED	A S o th 0
8	High	03 §7.1, §7.2 vs §4.2	`chat_template: gemma4` in both 31B recipes, while §4.2 requires training with thinking to match inference	S10 says use `gemma-4-thinking` for 31B/26B and `gemma-4` for the small models. As written, following the recipe produces no-thought training data — the exact failure mode §4.2 and 05 R7 warn about.	S10 (template choice), S8 (thought token format)	S g § g in v to
9	High	README §4 Path B vs §7; 03 §1, §9.4; 05 R6	"alpha 32 (scale 2.0) is a diagnostic only" (README §4) vs "Sweep alpha 8/16/32" (README §7, 03 §9.4) and "sweep 8/16/32 at inference" (03 §1)	Same corpus, opposite instruction. S25 has exactly one degenerate point at scale 2.0/rank 16, so recommending it as a sweep arm is a wasted submission cycle.	S25	S (a a n §
10	High	03 §3, §7.1–§7.4; 06 §4	Training-base IDs: `google/gemma-4-31B-it-qat-q4_0-unquantized`, `google/gemma-4-31B-it`, `google/gemma-4-31B`, `unsloth/gemma-4-31B-it`; 12B/E4B equivalents	S7/S43 give only the **suffix** convention (`-qat-q4_0-unquantized`); no ledger entry verifies an org/prefix for any training base. S8 verifies only the CT repo ID. 03:59 uses `…-31B-it` while 03:410/418 uses `google/gemma-4-31B` (which is what S12 uses), so the doc is internally inconsistent about the base it will download.	S12 (`google/gemma-4-31B`), S7/S43 (suffix only), S8 (CT ID)	P s 3 (r b u [
11	Medium	00 §7	"Top public score: 0.24 (four teams)"	S4 records the top score and the 0.22 cluster; no team count.	UNSUPPORTED (S4 has no count)	D v r
12	Medium	00 §8	"The public 0.24/0.22 notebooks are Apache 2.0 (S36, S37)"	Neither ledger entry records a license. Only S35 (repo) and S3 (winner license) are license facts. Readers are told to rely on a license that was never checked.	UNSUPPORTED; S35, S3	M th n S
13	Medium	00 §4; 02 §5.6; SOURCES S2	"256-d node embeddings"	S2's own entry reads "256 files each; … 256-d node embeddings" — the file count and the dimension appear side by side, which suggests conflation. No other source gives a dimension; S5:443 mentions only `.npz`.	S2 (ambiguous); UNSUPPORTED for the dimension	V th v o fr §
14	Medium	03 §5.3 vs §5.2	Rank-16 adapter "~110–220 MB … ~8 fit", while §5.2 says rank 16 is "~245 MB"	The same doc gives two sizes for the same adapter. 245 MB (S25, measured) implies ~12 fit under 3 GiB, not 8. The 110–220 MB row is S5's generic bf16 planning range, not this adapter.	S25 (245 MB measured), S5:198 (planning range)	S p (fo la r fr

#	Sev	File:section	Claim	Problem	Source that should be used / UNSUPPORTED	F
15	Medium	00 §9; README §4 Path C; 07 §2.1	"Best small-model public score was ~0.06 as of 2026-10-20 (S39/S47)"	No [COMMUNITY] label; the date and number live only in S47, which itself attributes them to S39 — but the S39 entry does not contain them. Circular citation to a single undated snapshot.	S47 (directly), S39 (does not contain it)	L S s s n
16	Medium	00 §7; 05 R11	"Identical files have scored 0.12–0.22 public (S34)"	One community study on 31 public tasks, stated as a fact about the leaderboard, without [COMMUNITY]. It is also used to justify the ±2-task noise band, which comes from a different source (S38/S39).	S34 ([COMMUNITY]), S38/S39 for the ±2-task band	L n c fo s
17	Medium	05 R16; 06 §1.2	"60/234 edit_file and 20/255 write_file calls rejected in one report (S30 thread)"	S30 is the queue-delay thread; its entry contains no tool-call rejection counts. 06 §1.2 cites "(S30, S39)" for malformed_calls, and neither entry has that data.	S30 (does not contain it); UNSUPPORTED	A to e 0
18	Medium	06 §2.3	"statements ≤120 chars had 7 resolves in 32 attempts vs 24 in 56 for longer statements (S39)"	Not in the S39 entry. The docs then use it to advise balancing statement length in the holdout.	S39 (does not contain it); UNSUPPORTED	A d b
19	Medium	02 §3.2	"a public comparison found 36 calls worse than 28 locally (S39)"	Not in the S39 entry (which records thinking-budget and 0.10/0.15 experiments, not call-budget A/Bs).	S39 (does not contain it); UNSUPPORTED	C d is S
20	Medium	02 §4.2; 04 §2.1	"The public 0.24 prompt is ~3,700 characters (S36)"; "use the public 0.24 prompt verbatim (S36)"	No character count in S36, and no local copy of that prompt exists in the workspace (the notebook writes the zip; the text is not captured in _research/). "Verbatim" cannot be checked offline.	S36 (no count, no local copy)	S S d c c n
21	Medium	04 §7.2	"loss (Gemma 4 31B loss is typically 1–3; S10)"	S10's entry records memory figures and get_peft_model args, not loss ranges.	S10 (does not contain it); UNSUPPORTED	R lo
22	Medium	05 R14	"LoRA runs hit pre-clip gradient norms 100–1000× neighbors (S11)"	S11 supports scaling=1.0 + QK-RMSNorm, LR 5e-5 and max_grad_norm 0.1; the 100–1000× figure is not in the entry.	S11 (partial); UNSUPPORTED for the magnitude	C th o w
23	Medium	05 R13; 03 §10.4	"Gemma 4 FSDP has reported FQN/gradient issues in other frameworks"; "Gemma 4 + FSDP/GRPO has reported instability"	No citation on either sentence. S14 (the cited support for R13) only lists which model types FSDP+QLoRA supports.	UNSUPPORTED (S14 does not cover it)	C is th c s
24	Medium	03 §5.2	"Rank 128 is the maximum"	S5:201 labels 128 max_lora_rank, but the notebook computes max(detected_ranks) with 128 only as the fallback and no clamp (notebook-facts.md, discrepancy 8). Treating 128 as	S5:195,201; S6:284; notebook-facts #8	R 1 d s

#	Sev	File:section	Claim	Problem	Source that should be used / UNSUPPORTED	F
				enforced is unsafe for anyone shipping a larger adapter.		
25	Medium	README §4 (opening); 04 §0	"4×A100 40 GB remote box plus Kaggle free GPUs" — no quota budget anywhere	S42 (weekly ~30 h quota; L4x4 consumes at 2× the T4/P2 rate) exists in the ledger and is cited by nothing. Any Kaggle-side plan (rehearsals, probe runs, E4B probe) silently assumes unlimited quota.	S42, S1	A R h L o c
26	Medium	00 §4	"The test set was filtered for solvability with a frontier model (S2)"	S2 says "a frontier-model **near-completion** filter" — a near-completion filter is not a solvability guarantee, and the docs elsewhere lean on "gold validates 100%" (S31) for that.	S2 (wording), S31	U th S s
27	Medium	03 §1 (decision table)	"Sequence length 8k–16k per-call samples … (S11, S13)"	Both cited sources are **32k** FSDP examples. They support "32k is possible on multi-GPU", not "8k–16k is the right choice".	S11, S13 (32k), S25 (max_length 32768)	C fe 1 [th to R
28	Low	02 §1.2	"a 12B coder + 12B reviewer solved 3/31 vs 4/31 for the coder alone (S34)" — "public evidence is mixed"	A 1-task difference at n=31 is noise by the docs' own rule (06 §3.4: "a 1-task win on 20 tasks is noise"). The reviewer run's budget is also not stated.	S34; 06 §3.4	S n b
29	Low	00 §9	"likely top several hundred"; "Winner | private 0.24+"	No source for the rank band or the private-score threshold; the table reads as measured.	UNSUPPORTED	D o [re c
30	Low	03 §6.3	"BF16 weights are ~61.4 GB total (S7)"	S7's 31B figure is 69.9 GB *inference memory*; 61.4 GB is 30.7B × 2 bytes (S8's parameter count). The derivation is right but the citation is not.	S8 (30.7B params), S7 (69.9 GB)	L B 6
31	Low	06 §4 (manifest)	`"vllm_version": "0.19.1"` as the reproducibility key	S39 documents that vLLM 0.19.1 was rebuilt twice **under the same version string**, so the version is not a build identity.	S39	A v in n c
32	Low	04 §1.2	`kaggle datasets download -d metric/gemma-4-developer-agent-wheelhouse`	The slug is inferred from a Kaggle *input path* (`S6:55`); the CLI itself is uncited although S41 exists. If the slug is wrong the command fails with a misleading 403.	S41 (CLI), S6:55 (path)	C [k - d
33	Low	README (Evidence rule)	Format examples "`[S5:123]`, `[S6:456]`"	They look like real citations and point at unrelated content (max YAML size; a tool response). A	—	U [

#	Sev	File:section	Claim	Problem	Source that should be used / UNSUPPORTED	F
				reader who spot-checks the evidence rule finds nonsense.		
34	Low	README §10	"Remove the sample dummy adapters unless trained"	S6:122 says the sample adapters are **31B-shaped**, not dummy; and deleting the files without deleting the `adapter:` keys breaks `validate_directory`/single-model checks.	S6:122	R (S a r / a
35	Low	README §5 (W5 row)	"entry/merger by Nov 24"	S1 says 2026-11-25. Reads as a date error rather than deliberate margin.	S1	" n 2
36	Low	03 §7.3	Unsloth snippet passes `max_seq_length` to both `FastModel.from_pretrained` and `SFTConfig`, and `tokenizer=` to `SFTTrainer`	The snippet is presented as runnable, but no ledger source shows this API shape (S10's entry records only `get_peft_model` args; S16's entry records trainer features, not argument names). The duplicated sequence-length argument is the kind of thing that fails on version drift.	S10 (partial), S16 (partial); `UNSUPPORTED` for the exact snippet	F in it " a th v v n
37	Low	07 §2.3	"It submits a patch on ~97% of tasks but often edits the wrong file"	The 97% is S34; "often edits the wrong file" is an inference from 0/31 solves (S34 gives no file-level data).	S34 (97%); second clause `[INFERENCE]`	S la
38	Low	03 §9.2	"With a rank-16 adapter … ~27,712 tokens at util 0.80 (S27). At 0.90 it may be higher, but still below the no-adapter baseline."	The second sentence is an unlabeled inference; S27 measured only at 0.80.	S27; second sentence `[INFERENCE]`	M a b
39	Low	00 §3/§4; 06 §2.1	"public tasks" / "public split" used for two different sets: the 129 training tasks and the ~58 LB tasks	00 §3 says "roughly 58 public tasks" while 00 §4 and 06 §2.1 say "129 public training tasks"; the same phrase names both.	S2 (129), S38/S39 (~58)	D " a u
40	Low	SOURCES	Orphan entries S9, S21, S41, S42, S44 are never cited in any doc	The ledger advertises coverage it does not provide; meanwhile the docs make claims those entries would support (vLLM LoRA serving, Kaggle CLI, quota, Unsloth QAT).	S21, S41, S42, S44, S9	C (S S in

Top 10 fixes

1. **Fix the 240 s / 300 s conflict (#2).** `evaluation.timeout_seconds` is Phase-1 only; Phase-2 pytest is fixed at 300 s (S5:506). Delete the "shorter timeout fails a correct patch" rationale in 02 §2.1/§2.3/§10 and flag S38's claim as contradicted by the official harness. This is the one finding that currently teaches a false mechanism.
2. **Collapse the budget guidance to one primary (#3).** 5 min / 24 calls / thinking 2048 everywhere; 8/28/4096 becomes "reference config, A-B only". Update README §9 and 02 §2/§10, which still ship the old default.
3. **Qualify the 0.24 result everywhere (#1).** S36 is author-reported, its sidebar shows 0.18, and the notebook does not run inference. 02 §2.3's rationale belongs to S38. The strategy is unchanged, but the premise must stop reading as verified.

4. **Correct the per-task score step (#4).** Final standings are over the private ~60, so one task ≈0.017, not 0.0083. This doubles the apparent noise floor of every decision rule in 06.
5. **Re-source or delete the S28 statistics (#5).** The 7-of-8 timeouts and `exit_loop` claims are not in S28; `exit_loop` belongs to S5:341. Structural advice resting on a fabricated statistic is worse than no statistic.
6. **Derive the LoRA serving parameters (#6).** `max_loras`/`max_lora_rank` come from the adapter map (S5:153). Hard-coding 16 makes the local A/B a different stack from the scorer for rank>16 adapters.
7. **Reconcile the S25 numbers (#7).** "Five temperatures", "12,602 tokens", "2.5k–14.7k chars" are the load-bearing evidence for the thinking-mismatch lesson and appear nowhere in the ledger; add quotes or remove them.
8. **Use `gemma-4-thinking` in the 31B recipes (#8).** As written, §7.1/§7.2 generate no-thought data while §4.2 requires thinking-matched data — the exact failure S25 documents.
9. **Unify the alpha sweep (#9).** 0.5/1.0 as candidates, 2.0 as a negative control only. README §4 and §7 currently disagree.
10. **Canonicalise the training-base IDs and verify them (#10).** `google/gemma-4-31B` vs `google/gemma-4-31B-it` vs `unsloth/gemma-4-31B-it`; none of the unquantized-QAT IDs is source-verified. One ID per size, with a ledger entry carrying URL + revision.

Ledger edits implied by this audit

Additions needed so the docs can stay honest without deleting useful detail:

- S2: split "256 files" from "256-d"; verify or drop the dimension.
- S25: add the five-temperature count, the 12,602-token template check, and the 2.5k–14.7k thought-length range, each with a quote.
- S30: add the tool-call rejection counts (60/234, 20/255) or state explicitly that the entry does not cover them.
- S36: add the prompt text (or its SHA-256 and length) and the license status; keep the author-reported caveat prominent.
- S39: add the ≤120-character statement statistic, or mark it as absent.
- S47: cite the small-model 0.06 snapshot to its own post, not to S39.
- New S50: canonical Hugging Face IDs + revisions for the training bases (31B/12B/E4B unquantized QAT and BF16).